

GENERATIVE ADVERSARIAL NETWORKS

“This (GANS), and the variations that are now being proposed is the most interesting idea in the last 10 years in ML, in my opinion”

-Yann LeCun

WHAT ARE GANS?

Generative Adversarial Networks

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Generative Adversarial Networks



Generative Models

We try to learn the underlying the distribution
from which our dataset comes from.

Eg: Variational AutoEncoders (VAE)

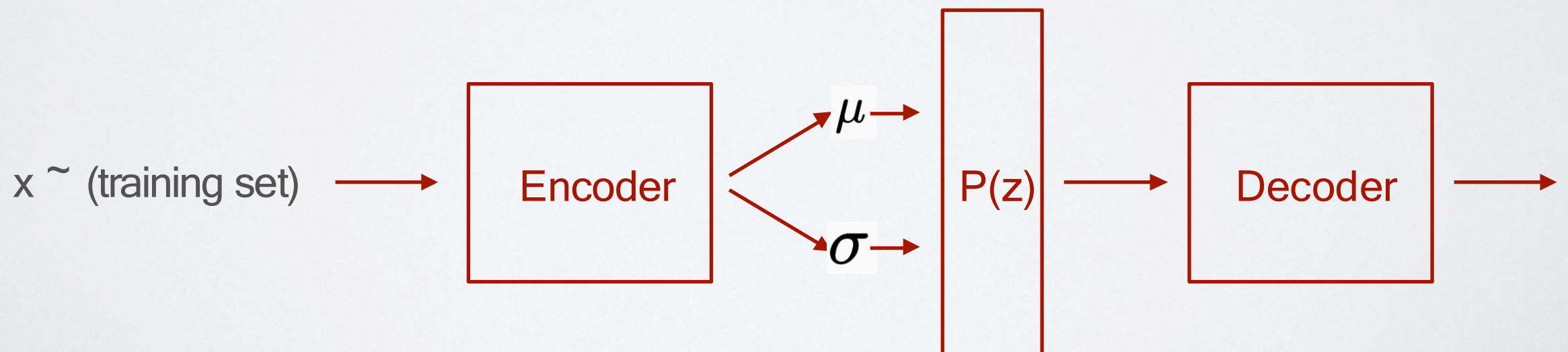
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Adversarial Training

GANS are made up of two competing networks (adversaries) that are trying beat each other.

WHAT ARE GANS?



Generative Models

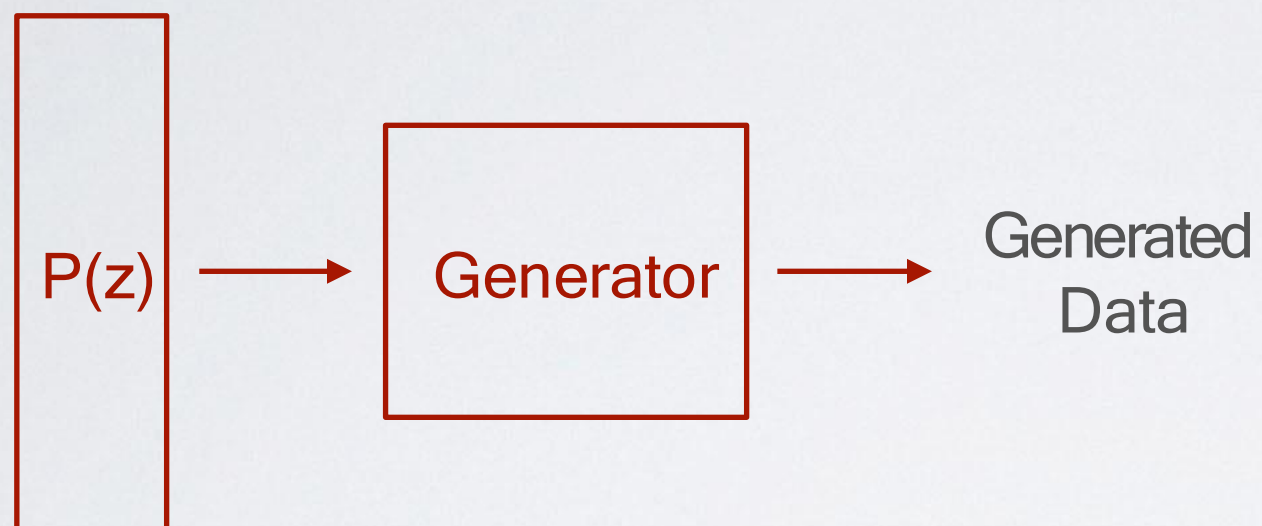
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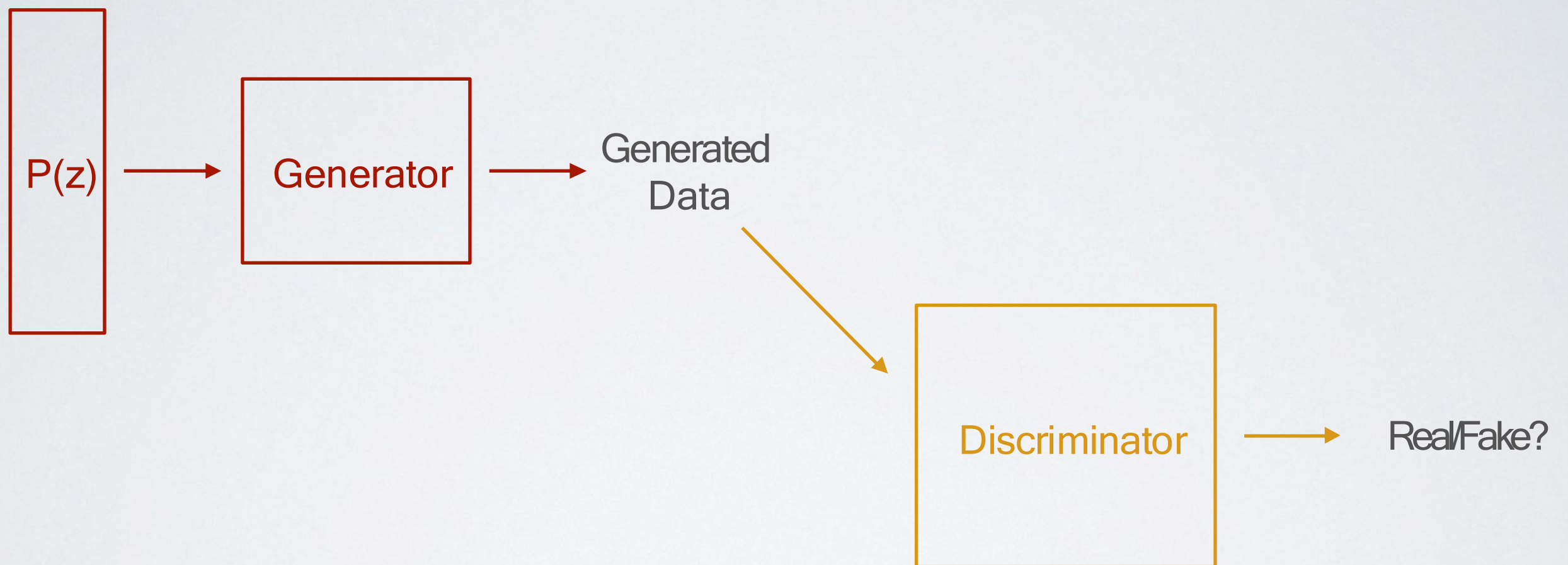
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Neural Networks

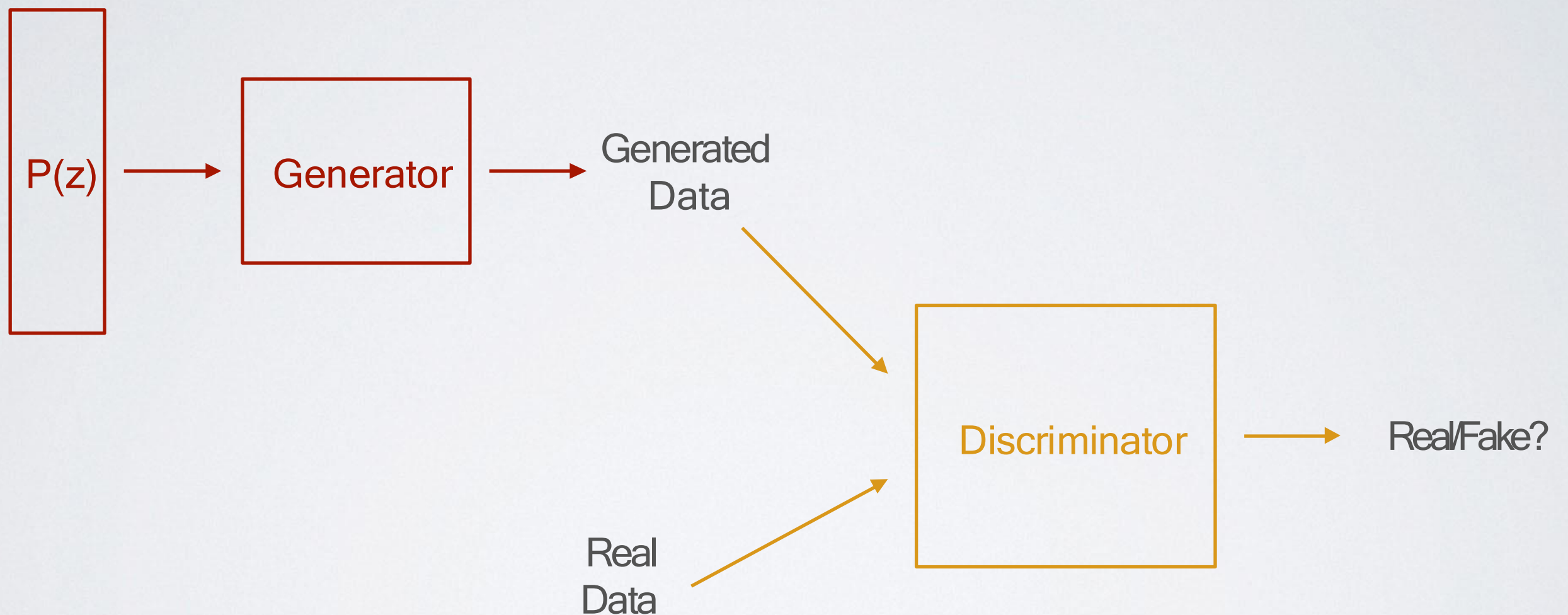
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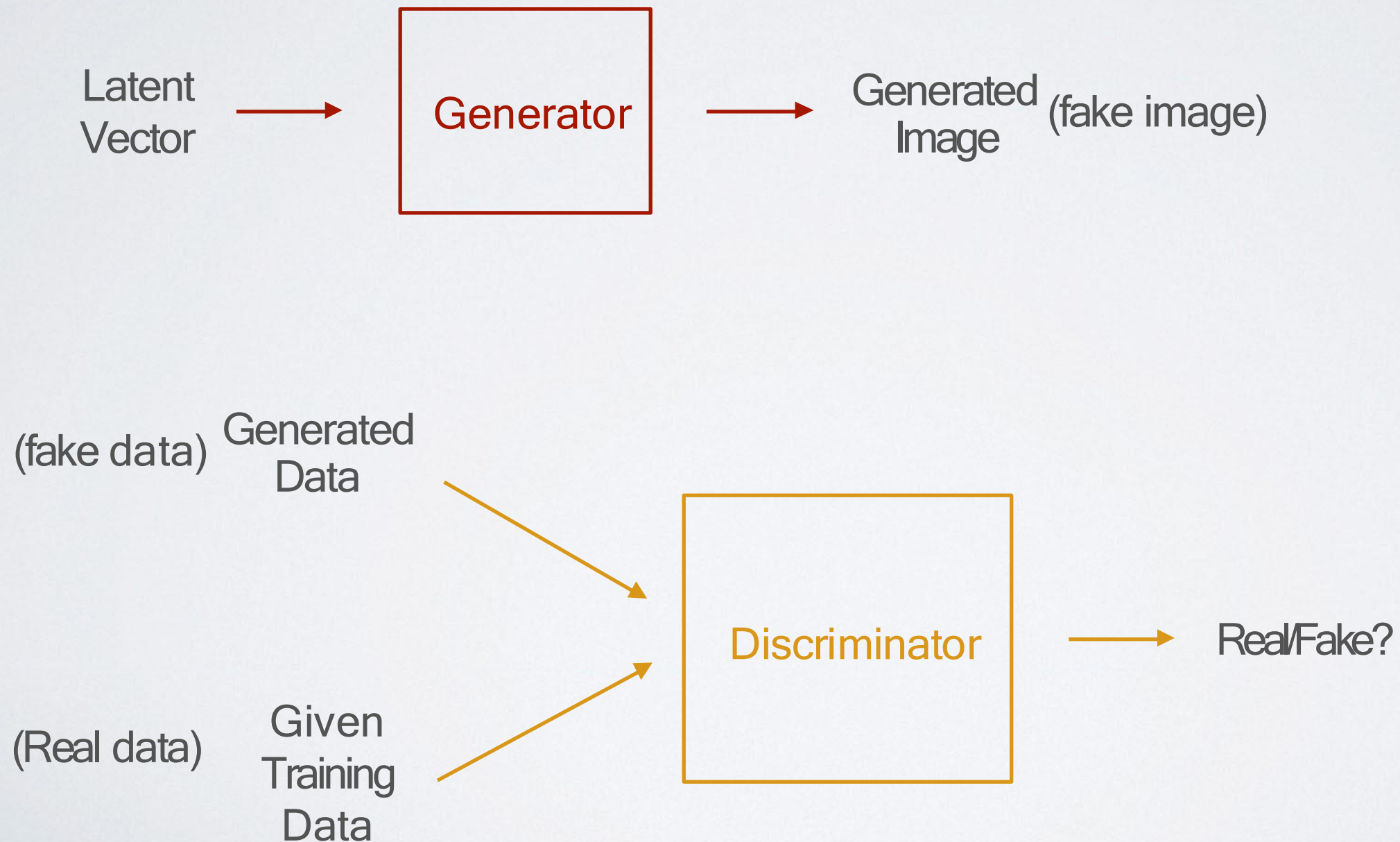
WHAT ARE GANS?



HOW TO TRAIN A GAN?

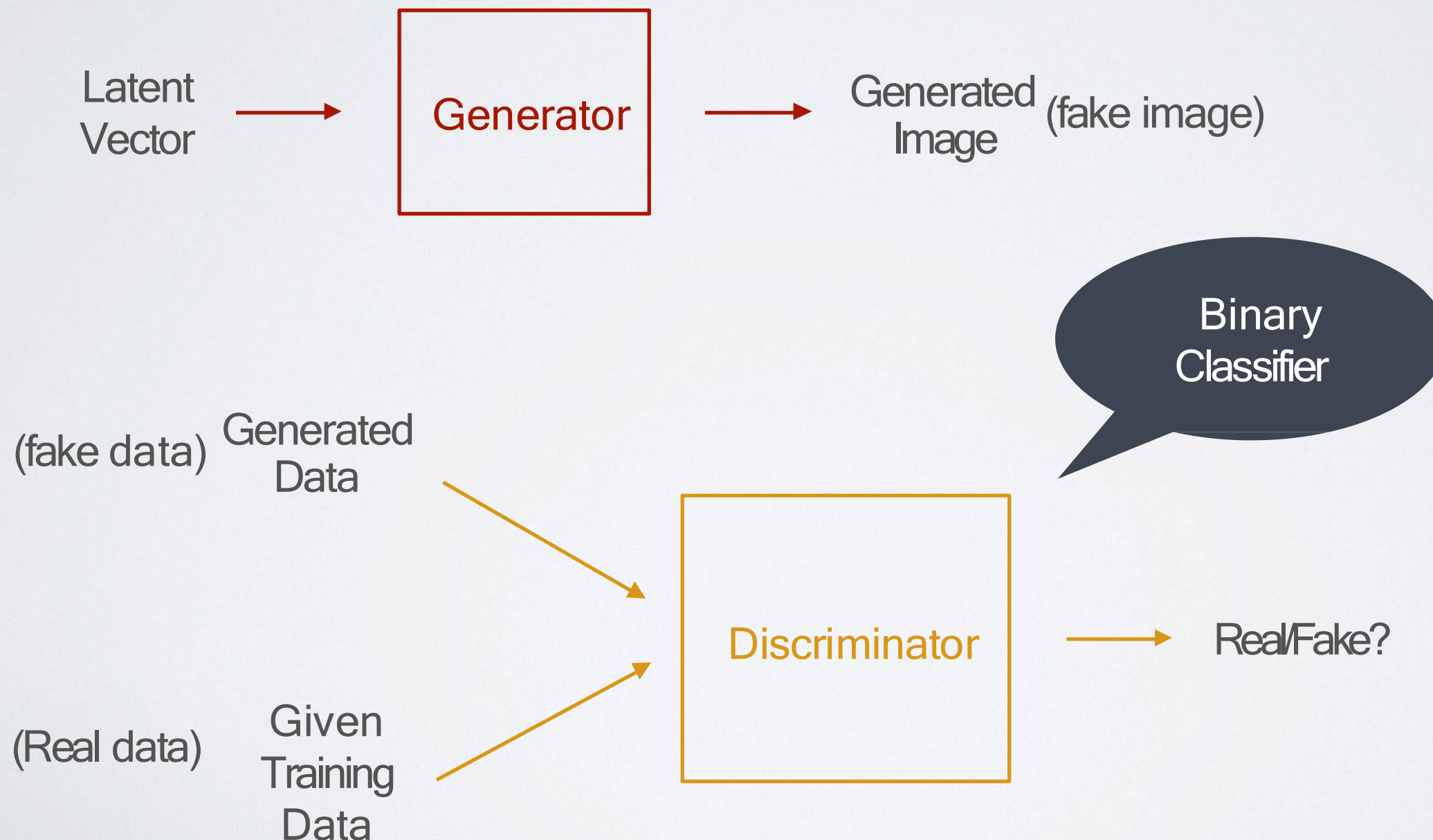
HOW TO TRAIN A GAN?

At $t = 0$,



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HOW TO TRAIN A GAN?

Which network should I train first?

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Discriminator!

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But with what training data?

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Discriminator!

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The Discriminator is a Binary classifier.

The Discriminator has two class - Real and Fake.

The data for Real class is already given: THE TRAINING DATA

The data for Fake class? -> generate from the Generator

HOW TO TRAIN A GAN?

What's next? -> Train the Generator

But how? What's our training objective?

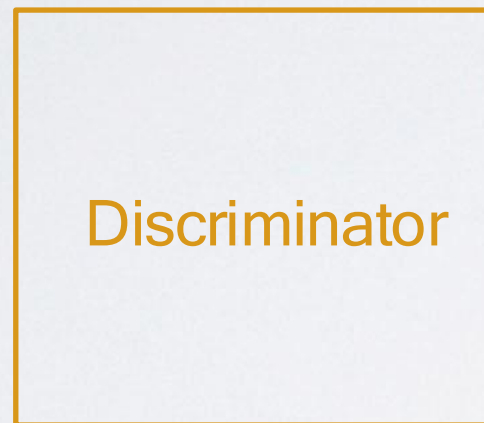
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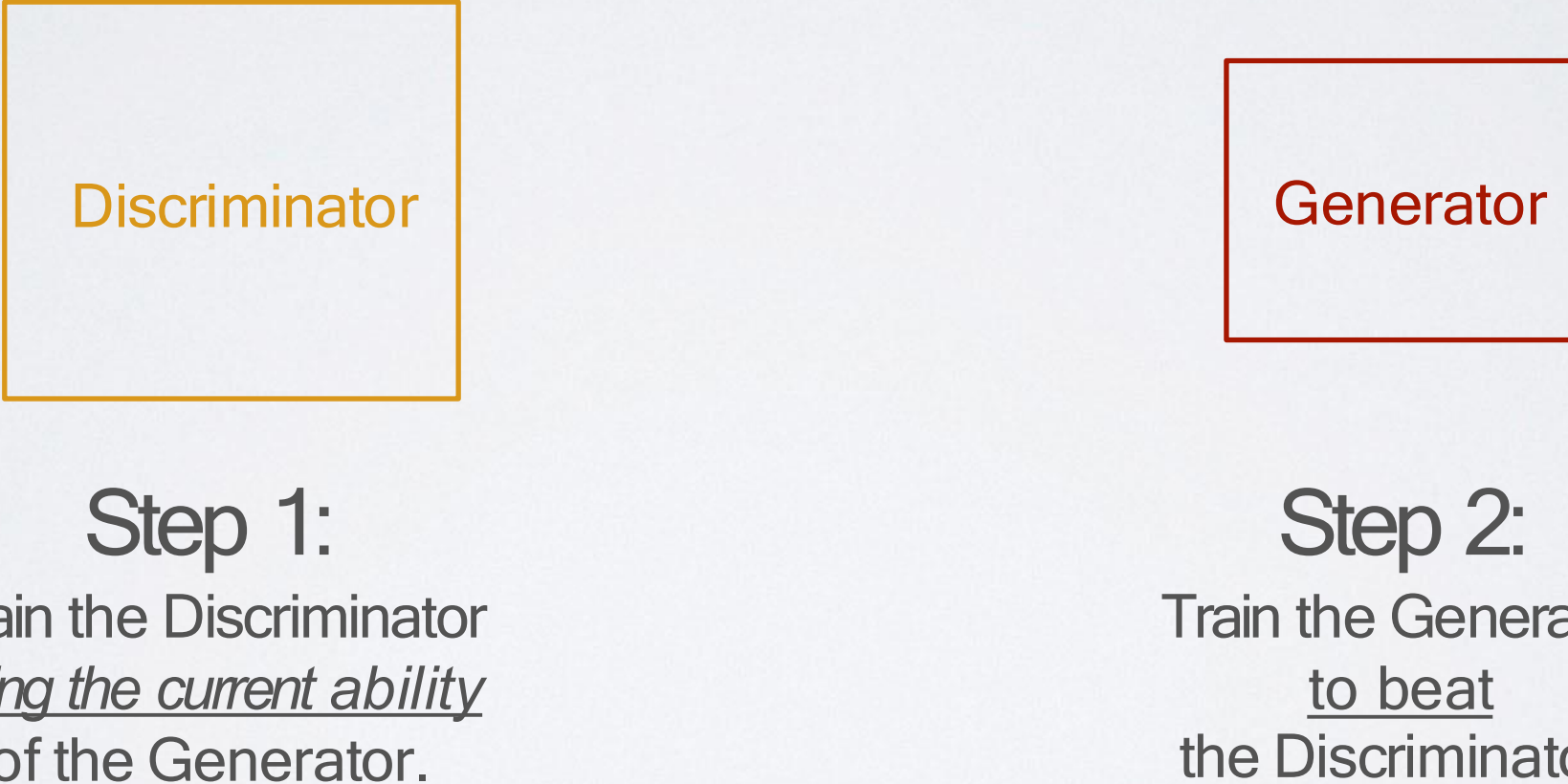
Generate images from the Generator
such that they are classified incorrectly by the Discriminator!

HOW TO TRAIN A GAN?



Step 1:
Train the Discriminator
using the current ability
of the Generator.

HOW TO TRAIN A GAN?



Discriminator

The diagram illustrates the two-step training process of a Generative Adversarial Network (GAN). It features two main components: a Discriminator and a Generator. The Discriminator is represented by an orange-outlined square on the left, and the Generator is represented by a red-outlined square on the right. Below each component, a step in the training process is described. Step 1, associated with the Discriminator, involves training it using the current ability of the Generator. Step 2, associated with the Generator, involves training it to beat the Discriminator. The text for Step 1 is in black, while the text for Step 2 is in gray.

Step 1:

Train the Discriminator
using the current ability
of the Generator.

Generator

Step 2:

Train the Generator
to beat
the Discriminator.

HOW TO TRAIN A GAN?



Discriminator

The diagram illustrates the two-step training process of a Generative Adversarial Network (GAN). It features two main components: a Discriminator, represented by an orange-outlined box on the left, and a Generator, represented by a red-outlined box on the right. Below the Discriminator box is 'Step 1: Train the Discriminator using the current ability of the Generator.' Below the Generator box is 'Step 2: Train the Generator to beat the Discriminator.' At the bottom center, a final instruction reads: 'Generate images from the Generator such that they are classified incorrectly by the Discriminator!'.

Step 1:

Train the Discriminator
using the current ability
of the Generator.

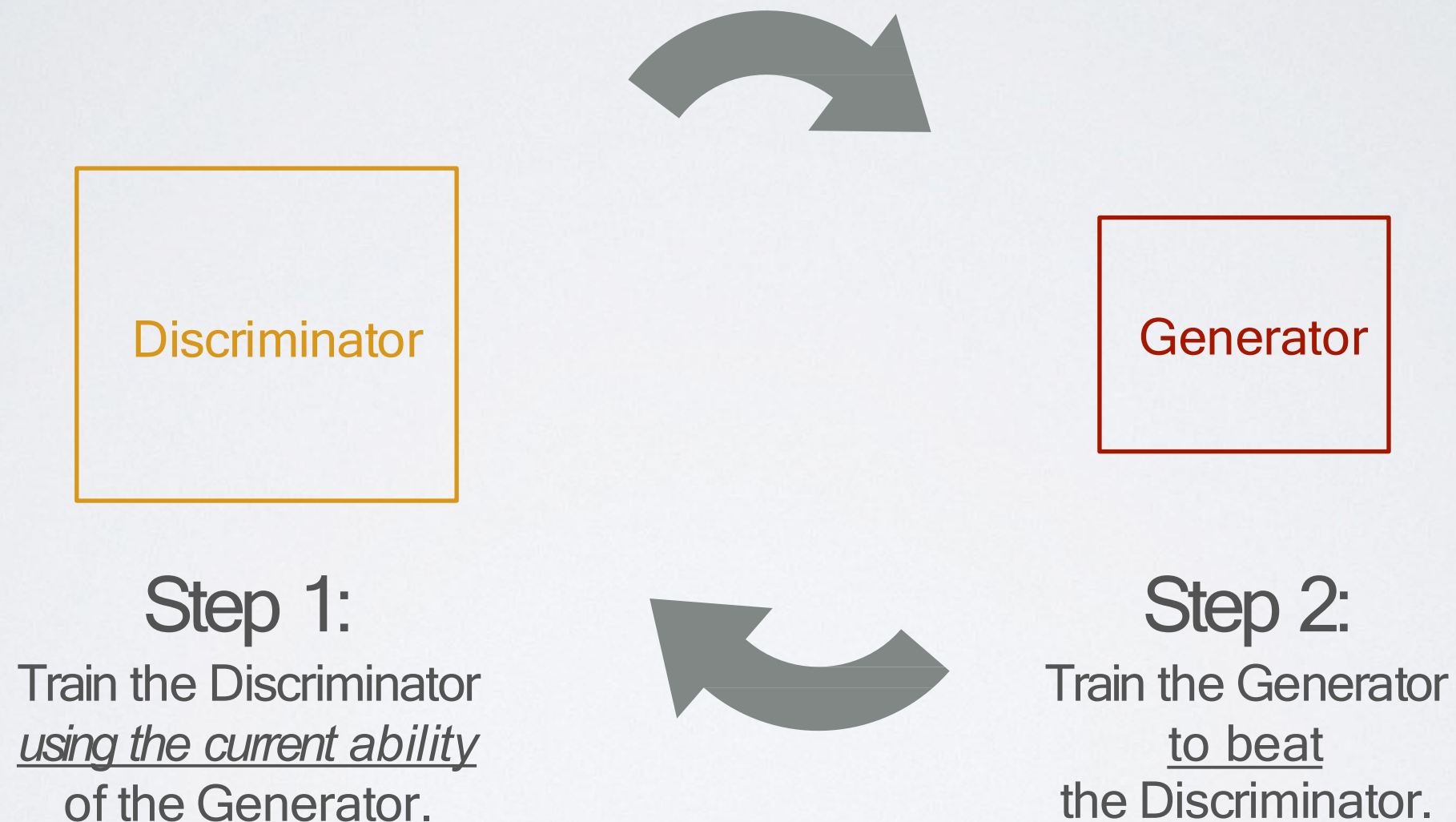
Generator

Step 2:

Train the Generator
to beat
the Discriminator.

Generate images from the Generator
such that they are classified incorrectly by the Discriminator!

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MNIST AND FASHION-MNIST

